Supplementary Material: Reference Requirements of the Benchmark

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This table lists every row and atom of the five reference sets (version 3). Rows are the published reference rows and rows added from the same reference document (marked †); each row is followed by its atoms, the scoring units. Row statements are curator paraphrases. Atom statements are written in one standard form: a content atom says what users can know, find, access, cite or use (*Users can ...*), a profile atom what the profile does (*The profile ...*), and a system atom names the system or role the source names. The reference documents' own wording is cited in the benchmark files. A provisional adjudication of reviewed items was applied on 23 October 2026: one duplicate merged, one umbrella atom left out of scoring, two compound atoms split, three evidence labels downgraded and one omitted need restored; each change is logged in the benchmark files.

Columns. Lev.: the atom's type: content (c), information a catalog record should carry; profile (p), how the profile is specified (principles, scope, reuse and alignment, obligation levels, code lists, documentation); or system (s), the systems and processes around the profile (governance and maintenance, validation, interfaces, exchange, review)[citation needed]. Only content atoms are scored. *Status* in the source: requirement (req.), proposal (prop.), open issue (open), guidance (guid.) or deferred to future work (def.). *Base*: relation to the pinned DCAT-AP release: restates, tightens, extends, partly covered by, or beyond the base; – for atoms that are not about metadata terms, ? if unresolved. *Raw* and *Cons.*: whether the raw corpus and, for mobilityDCAT-AP and HealthDCAT-AP, the pre-consolidation corpus state the atom: S stated, P in part, U uncertain, N not stated, – no such corpus. These are AI labels with verbatim quotations checked by program; human review is pending. *Sc.*: scored by default (✓); m merged into an equivalent atom of another row (`Req-24.1` into `Req-19.7`); u an umbrella of separately scored atoms (`H-4.1` of `H-16.1`, `H-26.1`, `H-28.1`, `H-36.1`, `H-66.2`).

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
mobilityDCAT-AP NAPCORE requirement analysis (2022), Table 1; base DCAT-AP 2.0.1; test case 40 rows (40 published), 98 atoms, 39 scored (39 requirements), 7 restate the base, 31 stated (37 at least in part); pre-consolidation corpus: 38 (39)							
`Req-1` Use sufficiently expressive, context-aware and modular descriptions with explicit structure, types and rules; distinguish entity identities, preserve references, and support persistent identifiers. *(Table 1, PDF pp. 15–19)*							
`Req-1.1`	The profile makes metadata descriptions expressive.	p	req.	–	S	S	
`Req-1.2`	The profile makes metadata descriptions rich.	p	req.	–	S	U	
`Req-1.3`	The profile describes the data adequately.	p	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
`Req-1.4`	The profile supports easy access to the data.	p	req.	–	S	S	
`Req-1.5`	The profile makes metadata descriptions context-sensitive.	p	req.	–	U	S	
`Req-1.6`	The profile makes metadata descriptions modular.	p	req.	–	S	S	
`Req-1.7`	The profile gives metadata descriptions a detailed structure.	p	req.	–	S	S	
`Req-1.8`	The profile uses strong typing and rules in metadata descriptions.	p	req.	–	S	S	
`Req-1.9`	The profile keeps referential integrity.	p	req.	–	S	U	
`Req-1.10`	The profile enables persistent identifiers.	p	req.	–	S	S	
`Req-2` Make data findable and reusable; provide a basis for NAP metadata databases, describe catalogues, data and their delivery options, and support metadata federation. *(Table 1, PDF pp. 15–19)*							
`Req-2.1`	The profile makes datasets discoverable and searchable.	p	req.	–	S	S	
`Req-2.2`	The profile makes datasets reusable.	p	req.	–	S	S	
`Req-2.3`	The profile serves as a baseline for developing a NAP metadata database.	p	req.	–	S	S	
`Req-2.4`	The profile allows a detailed description of the registry (catalogue).	p	req.	–	S	S	
`Req-2.5`	The profile allows a detailed description of the registry's datasets.	p	req.	–	S	S	
`Req-2.6`	The profile allows a detailed description of the datasets' distributions and/or data services.	p	req.	–	S	S	
`Req-2.7`	The profile allows aggregation and exchange of metadata.	p	req.	–	S	S	
`Req-3` Expose descriptions in forms suitable for people and software. *(Table 1, PDF pp. 15–19)*							
`Req-3.1`	The profile provides for metadata in a human-readable format.	p	req.	–	S	S	
`Req-3.2`	The profile provides for metadata in a machine-readable format.	p	req.	–	S	S	
`Req-4` Accommodate descriptions in several languages. *(Table 1, PDF pp. 15–19)*							
`Req-4.1`	The profile provides for metadata information in multiple languages.	p	req.	–	S	S	
`Req-5` Use shared vocabularies so participants interpret metadata consistently. *(Table 1, PDF pp. 15–19)*							
`Req-5.1`	The profile adopts a standardised set of vocabularies to ensure semantic interoperability.	p	req.	–	S	S	
`Req-6` Use a common serialization that software agents can process automatically. *(Table 1, PDF pp. 15–19)*							
`Req-6.1`	The profile adopts a standard format or serialisation, for technological interoperability and automated access by software agents.	p	req.	–	S	S	
`Req-7` Publish freely accessible documentation on the Web and, where feasible, provide digitally annotated repository documentation. *(Table 1, PDF pp. 15–19)*							
`Req-7.1`	The profile's documentation is publicly available on the Web without proprietary licences.	s	req.	–	P	S	
`Req-7.2`	The profile's documentation is annotated digitally in a public repository, where possible.	s	req.	–	P	P	
`Req-8` Establish governance and revision procedures, public issue discussion and release planning, and incorporate resolved changes in later versions. *(Table 1, PDF pp. 15–19)*							
`Req-8.1`	The profile defines the governance processes for its maintenance.	s	req.	–	S	S	
`Req-8.2`	The profile defines its revision methodology.	s	req.	–	S	S	
`Req-8.3`	The profile allows public commenting on its development.	s	req.	–	S	S	
`Req-8.4`	The profile has an issues and releases planning tool.	s	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-8.5	The profile's authors discuss and resolve issues and proposals and incorporate them into follow-up releases.	s	req.	–	S	S	
Req-9 Build on DCAT-AP terms when suitable and follow its extension rules for additions. *(Table 1, PDF pp. 15–19)*							
Req-9.1	The profile reuses DCAT-AP classes and properties where possible.	p	req.	–	S	S	
Req-9.2	The profile adds any further classes and properties according to the DCAT-AP extension rules.	p	req.	–	S	S	
Req-10 Account for subsequent changes to the DCAT-AP baseline. *(Table 1, PDF pp. 15–19)*							
Req-10.1	The profile follows the ongoing evolution of DCAT-AP, considering its developments and revisions.	p	req.	–	S	S	
Req-11 Review the complete CMC element set against mobility catalogue needs. *(Table 1, PDF pp. 15–19)*							
Req-11.1	The profile considers all CMC metadata elements, to identify metadata that better fit the specific requirements of NAPs.	p	req.	–	S	S	
Req-12 Account for other DCAT-AP profiles and INSPIRE metadata practice. *(Table 1, PDF pp. 15–19)*							
Req-12.1	The profile considers existing DCAT-AP extensions (e.g. GeoDCAT-AP for geospatial data).	p	req.	–	S	S	
Req-12.2	The profile considers the metadata specifications of the INSPIRE Directive.	p	req.	–	S	S	
Req-13 Preserve interoperability with code lists already deployed by NAP operators. *(Table 1, PDF pp. 15–19)*							
Req-13.1	The profile takes into account the vocabularies already used in operational NAPs when defining controlled vocabularies, for interoperability with existing solutions.	p	req.	–	P	S	
Req-14 Describe the varied mobility domain in enough detail to distinguish its growing dataset inventory. *(Table 1, PDF pp. 15–19)*							
Req-14.1	The profile provides exhaustive metadata descriptions for mobility data, given the growing number and variety of mobility datasets.	p	req.	–	P	S	
Req-15 Define a common essential element set while permitting deployment-specific additions. *(Table 1, PDF pp. 15–19)*							
Req-15.1	The profile proposes essential metadata elements relevant to any NAP, as a minimum set of metadata for European NAPs.	p	req.	–	S	S	
Req-15.2	The profile allows further metadata elements in individual NAP deployments.	p	req.	–	S	S	
Req-16 Examine the established CMC/DCAT-AP correspondences to identify unrepresented information. *(Table 1, PDF pp. 15–19)*							
Req-16.1	The profile uses the canonical mappings between DCAT-AP and the CMC to identify information that DCAT-AP cannot yet represent.	p	req.	–	P	S	
Req-17 Record creation and revision times, description language, and responsibility for maintaining a metadata record. *(Table 1, PDF pp. 15–19)*							
Req-17.1	Users can know the date and time a metadata record was created or last modified.	c	req.	restates	S	S	✓
Req-17.2	Users can know the language in which the metadata are written.	c	req.	tightens	S	S	✓
Req-17.3	Users can know who is responsible for creating and maintaining the metadata.	c	req.	extends	S	S	✓
Req-18 State how the metadata record itself may be reused. *(Table 1, PDF pp. 15–19)*							
Req-18.1	Users can know the conditions of use of the metadata itself.	c	req.	restates	S	S	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-19 Characterise the data using a title and synopsis, resource kind, dataset and service classifications including MMTIS service categories, language and payload georeferencing. *(Table 1, PDF pp. 15–19)*							
Req-19.1	Users can know the name of a resource.	c	req.	restates	S	S	✓
Req-19.2	Users can know a brief description of a resource.	c	req.	restates	S	S	✓
Req-19.3	Users can know the type of a resource (dataset or service).	c	req.	restates	S	S	✓
Req-19.4	Users can know the dataset type category, by logical clustering or by the EC Delegated Regulations.	c	req.	beyond	S	S	✓
Req-19.5	Users can know the service type category listed in Delegated Regulation (EU) 2017/1926.	c	req.	beyond	P	P	✓
Req-19.6	Users can know the language of a resource.	c	req.	restates	S	S	✓
Req-19.7	Users can know the georeferencing method applied within the payload.	c	req.	beyond	S	S	✓
Req-20 Indicate which ITS delegated regulations apply to a resource. *(Table 1, PDF pp. 15–19)*							
Req-20.1	Users can know which Delegated Regulation (MMTIS, RTTI, SRTI, SSTP) covers a dataset or service.	c	req.	beyond	P	S	✓
Req-21 Coordinate, host, maintain and extend multilingual code lists for describing dataset and service content. *(Table 1, PDF pp. 15–19)*							
Req-21.1	The profile supports the coordinated development of a controlled vocabulary for describing dataset and service content.	p	req.	–	S	S	
Req-21.2	The profile hosts the controlled vocabulary.	p	req.	–	P	S	
Req-21.3	The profile keeps the controlled vocabulary maintained.	p	req.	–	S	S	
Req-21.4	The profile keeps the controlled vocabulary extendable.	p	req.	–	S	S	
Req-21.5	The profile allows the controlled vocabulary to be translated into local languages, for harmonisation.	p	req.	–	S	S	
Req-22 Represent the interval over which the published data remain valid. *(Table 1, PDF pp. 15–19)*							
Req-22.1	Users can know the period in which the published data do not expire.	c	req.	extends	S	S	✓
Req-23 Describe spatial extent, applicable regions and network coverage using established functional classifications. *(Table 1, PDF pp. 15–19)*							
Req-23.1	Users can know the area a dataset covers.	c	req.	tightens	S	S	✓
Req-23.2	Users can know the regions in which a dataset is valid.	c	req.	tightens	S	S	✓
Req-23.3	Users can know the transport network a dataset considers, following existing, widely applicable functional classifications.	c	req.	beyond	P	S	✓
Req-24 Identify payload location-reference methods and coordinate reference systems. *(Table 1, PDF pp. 15–19)*							
Req-24.1	Users can know the location referencing methods used in a dataset or service.	c	req.	beyond	S	S	m
Req-24.2	Users can know the coordinate reference systems used in a dataset or service.	c	req.	tightens	S	S	✓
Req-25 Identify the transport modes and operators represented by the data. *(Table 1, PDF pp. 15–19)*							
Req-25.1	Users can know the transport modes a dataset deals with.	c	req.	beyond	S	S	✓
Req-25.2	Users can know the transport operators a dataset deals with.	c	req.	beyond	S	S	✓
Req-26 Identify the publishing and owning parties for a dataset. *(Table 1, PDF pp. 15–19)*							
Req-26.1	Users can know who publishes a dataset.	c	req.	tightens	S	S	✓
Req-26.2	Users can know who owns a dataset.	c	req.	beyond	S	S	✓
Req-27 Describe contractual, licensing and other conditions on data use. *(Table 1, PDF pp. 15–19)*							
Req-27.1	Users can know whether a licence governs the use of the data.	c	req.	tightens	S	S	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-27.2	Users can know whether a contract governs the use of the data.	c	req.	extends	S	S	✓
Req-27.3	Users can know any other condition for using the data.	c	req.	extends	S	S	✓
Req-28	Describe the encoding, syntax, grammar and underlying data model used for access. *(Table 1, PDF pp. 15–19)*						
Req-28.1	Users can know the encoding of the data for access.	c	req.	beyond	S	S	✓
Req-28.2	Users can know the syntax of the data.	c	req.	tightens	S	S	✓
Req-28.3	Users can know the grammar of the data.	c	req.	beyond	S	S	✓
Req-28.4	Users can know the data model of the data.	c	req.	beyond	S	S	✓
Req-29	Offer schema information and example data so consumers can assess suitability before integration. *(Table 1, PDF pp. 15–19)*						
Req-29.1	Users can know the data schema, so they can evaluate the data before setting up consumption.	c	req.	beyond	S	S	✓
Req-29.2	Users can access data samples, so they can evaluate the data before setting up consumption.	c	req.	extends	S	S	✓
Req-30	Distinguish precise data-model versions, publication types and extensions, such as those used in DATEX II, for discovery. *(Table 1, PDF pp. 15–19)*						
Req-30.1	Users can know precisely the version of the data model a dataset adopts (e.g. the DATEX II version), to allow better searching.	c	req.	beyond	P	S	✓
Req-30.2	Users can know the publication type of the data model a dataset adopts (e.g. the DATEX II publication type), to allow better searching.	c	req.	beyond	U	S	✓
Req-30.3	Users can know the extensions of the data model a dataset adopts (e.g. DATEX II extensions), to allow better searching.	c	req.	beyond	U	S	✓
Req-31	Record update cadence, quality criteria and the history and status of ITS-regulation compliance assessments. *(Table 1, PDF pp. 15–19)*						
Req-31.1	Users can know the update rate of a dataset.	c	req.	tightens	S	S	✓
Req-31.2	Users can know the quality criteria of a dataset.	c	req.	beyond	S	S	✓
Req-31.3	Users can know the history of the procedures that assess compliance with the Delegated Regulations on providing data via a NAP.	c	req.	beyond	P	S	✓
Req-31.4	Users can know the status of the procedures that assess compliance with the Delegated Regulations on providing data via a NAP.	c	req.	beyond	S	S	✓
Req-32	Define occurrence limits and obligation levels and explain their consequences for database storage, user interfaces and harvesting, consistently with DCAT-AP. *(Table 1, PDF pp. 15–19)*						
Req-32.1	The profile specifies the cardinality of each metadata element.	p	req.	–	S	S	
Req-32.2	The profile specifies the obligation of each metadata element (mandatory, recommended, optional).	p	req.	–	S	S	
Req-32.3	The profile establishes clear rules on the cardinalities of its metadata elements for NAP deployments (e.g. their implications for database set-ups, GUI designs and API or harvesting mechanisms).	p	req.	–	P	S	
Req-32.4	The profile considers the existing DCAT-AP rules on cardinalities.	p	req.	–	S	S	
Req-33	Provide identifier guidance. *(Table 1, PDF pp. 15–19)*						
Req-33.1	The profile defines guidelines for identifiers.	p	req.	–	S	S	
Req-34	Define permitted values for metadata elements, using suitable controlled vocabularies. *(Table 1, PDF pp. 15–19)*						
Req-34.1	The profile defines the expected value range of each metadata element (for example with controlled vocabularies).	p	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
`Req-35` Track revisions and versions of catalogue metadata. *(Table 1, PDF pp. 15–19)*							
`Req-35.1`	Users can know the changes of metadata records on a NAP.	c	req.	restates	S	S	✓
`Req-35.2`	Users can know the different versions of metadata records on a NAP.	c	req.	extends	P	S	✓
`Req-36` Enable automated conformance checking of metadata. *(Table 1, PDF pp. 15–19)*							
`Req-36.1`	The profile supports the validation, possibly automatic, of metadata that adopts it.	s	req.	–	S	S	
`Req-37` Explain element meanings and how to apply them within NAP operations. *(Table 1, PDF pp. 15–19)*							
`Req-37.1`	The profile provides usage rules in addition to the metadata definitions, explaining the semantics of each metadata element.	p	req.	–	S	S	
`Req-37.2`	The profile advises how to relate the metadata elements to NAP processes.	p	req.	–	P	S	
`Req-38` Explain allocation of metadata between a dataset or service and its multiple delivery resources. *(Table 1, PDF pp. 15–19)*							
`Req-38.1`	The profile provides guidelines and solutions for dividing metadata between a dataset or service and its multiple resources.	p	req.	–	S	S	
`Req-39` Explain exchange of mobility-specific descriptions with other-domain portals, including handling extensions during harvesting. *(Table 1, PDF pp. 15–19)*							
`Req-39.1`	The profile provides guidelines on making NAP-specific metadata interoperable with other data portals (e.g. outside the mobility domain).	s	req.	–	S	S	
`Req-39.2`	The profile's interoperability guidelines cover how napDCAT-AP extensions are handled when harvesting from other portals.	s	req.	–	P	S	
`Req-40` Give guidance for assuring the quality of metadata received by a NAP. *(Table 1, PDF pp. 15–19)*							
`Req-40.1`	The profile provides guidelines for a quality assurance process that checks the quality of metadata collected on a NAP.	s	req.	–	S	S	

GeoDCAT-AP GeoDCAT-AP 1.0 (2015); base DCAT-AP 1.1; test case

46 rows (7 published), 56 atoms, 37 scored (37 requirements), 16 restate the base, 37 stated (37 at least in part)

`G-1` Provide RDF representations of spatial metadata that conform to DCAT-AP, retain meaning as far as practical and support cross-domain reuse. *(GeoDCAT-AP 1.0 §1.1;4.1)*							
`G-1.1`	The profile provides a DCAT-AP-conformant representation of geospatial metadata.	p	req.	–	P	–	
`G-1.2`	The profile provides an RDF representation of geospatial metadata that is as comprehensive as possible and avoids semantic loss.	p	req.	–	S	–	
`G-1.3`	The profile promotes the cross-domain reuse of geospatial metadata.	p	req.	–	P	–	
`G-2` Include every mandatory baseline DCAT-AP element. *(GeoDCAT-AP 1.0 §4.1)*							
`G-2.1`	The profile includes, at least, all mandatory DCAT-AP elements.	p	req.	–	U	–	
`G-3` Use other vocabularies for geospatial information only where the baseline does not cover it. *(GeoDCAT-AP 1.0 §4.1)*							
`G-3.1`	The profile uses vocabularies other than DCAT-AP only for geospatial metadata elements that DCAT-AP does not support.	p	req.	–	P	–	
`G-4` Base alignments on established practice and working-group implementation experience. *(GeoDCAT-AP 1.0 §4.1)*							
`G-4.1`	The profile bases its alignments as far as possible on existing practice, in particular the working group's experience.	p	req.	–	P	–	
`G-4.2`	The profile avoids a negative impact on existing implementations.	p	req.	–	U	–	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-5 Assess any new terms carefully and avoid conflicting with ongoing standardisation. *(GeoDCAT-AP 1.0 §4.1)*							
G-5.1	The profile defines new terms only when existing vocabularies offer no suitable candidate, and only after careful assessment.	p	req.	–	P	–	
G-5.2	The profile discards new terms that could conflict with standards under preparation.	p	req.	–	N	–	
G-6 Cover the ISO core and common INSPIRE metadata elements; exclude only optional material where necessary, with theme-specific extensions outside the common scope. *(GeoDCAT-AP 1.0 §4.2)*							
G-6.1	The profile covers all metadata elements of the core profile of ISO 19115.	p	req.	–	S	–	
G-6.2	The profile covers the INSPIRE metadata elements, leaving out only those not common to all INSPIRE spatial data themes.	p	req.	–	P	–	
G-6.3	The profile excludes only optional elements.	p	req.	–	U	–	
G-7 Offer an additional RDF publication route while retaining the existing INSPIRE metadata obligations and implementations. *(GeoDCAT-AP 1.0 §1.1)*							
G-7.1	The profile offers an RDF syntax binding as an additional way to implement and publish INSPIRE/ISO metadata on DCAT-AP-conformant portals.	p	req.	–	U	–	
G-7.2	The profile leaves the INSPIRE metadata regulation and technical guidelines, and the collections maintained under them, unchanged.	p	req.	–	N	–	
G-8† Identify who is responsible for a metadata record and how to contact them. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.16, II.20)*							
G-8.1	Users can know who is responsible for a metadata record and how to contact them.	c	req.	extends	S	–	✓
G-9† Record the date of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11, II.21)*							
G-9.1	Users can know the date of each metadata record.	c	req.	restates	S	–	✓
G-10† State the language of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.7, II.22)*							
G-10.1	Users can know the language of each metadata record.	c	req.	restates	S	–	✓
G-11† State the character set of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.19, II.24)*							
G-11.1	Users can know the character set of a metadata record.	c	req.	beyond	S	–	✓
G-12† Give each metadata record its own identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.17)*							
G-12.1	Users can know the file identifier of a metadata record.	c	req.	extends	S	–	✓
G-13† State the metadata standard a record follows. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.18)*							
G-13.1	Users can know the metadata standard a record follows.	c	req.	restates	S	–	✓
G-14† State the version of the metadata standard a record follows. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.18)*							
G-14.1	Users can know the version of the metadata standard a record follows.	c	req.	extends	S	–	✓
G-15† Give each resource a title. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.1)*							
G-15.1	Users can know the title of each resource.	c	req.	restates	S	–	✓
G-16† Give the dates of creation, publication and last revision of a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11)*							
G-16.1	Users can know the date of creation of a resource.	c	req.	beyond	S	–	✓
G-16.2	Users can know the date of publication of a resource.	c	req.	restates	S	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-16.3	Users can know the date of last revision of a resource.	c	req.	restates	S	–	✓
G-17†	Describe each resource in a short abstract. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.2)*						
G-17.1	Users can know a short abstract describing each resource.	c	req.	restates	S	–	✓
G-18†	State the language or languages used in a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.7)*						
G-18.1	Users can know the language or languages used in a resource.	c	req.	restates	S	–	✓
G-19†	Classify each resource by topic category. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.8)*						
G-19.1	Users can know the topic category of each resource.	c	req.	beyond	S	–	✓
G-20†	Give the geographic extent of a resource as a bounding box or a geographic identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.10)*						
G-20.1	Users can know the geographic extent of a resource, as a bounding box or a geographic identifier.	c	req.	restates	S	–	✓
G-21†	State the character encoding of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.24)*						
G-21.1	Users can know the character encoding of a dataset.	c	req.	beyond	S	–	✓
G-22†	Give the temporal extent a resource covers. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11)*						
G-22.1	Users can know the temporal extent a resource covers.	c	req.	restates	S	–	✓
G-23†	Describe the lineage of a dataset: how it was produced and processed. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.12)*						
G-23.1	Users can know the lineage of a dataset: how it was produced and processed.	c	req.	restates	S	–	✓
G-24†	State how spatial data are represented, e.g. as vector or grid data. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.26)*						
G-24.1	Users can know how spatial data are represented, e.g. as vector or grid data.	c	req.	beyond	S	–	✓
G-25†	State the format in which each distribution is encoded. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.25)*						
G-25.1	Users can know the format in which each distribution is encoded.	c	req.	restates	S	–	✓
G-26†	State the spatial resolution of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.13)*						
G-26.1	Users can know the spatial resolution of a dataset.	c	req.	beyond	S	–	✓
G-27†	Identify the organisations responsible for a resource and the role each plays. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.16)*						
G-27.1	Users can know the organisations responsible for a resource and the role each plays.	c	req.	partly	S	–	✓
G-28†	Give links to the resource or to further information about it. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.4)*						
G-28.1	Users can access the resource, or further information about it, through links.	c	req.	restates	S	–	✓
G-29†	State the coordinate reference system of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.23)*						
G-29.1	Users can know the coordinate reference system of a dataset.	c	req.	extends	S	–	✓
G-30†	State the temporal reference system of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.23)*						
G-30.1	Users can know the temporal reference system of a dataset.	c	req.	extends	S	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-31†	State the specifications a resource conforms to, and whether it conforms. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14)*						
G-31.1	Users can know the specifications a resource conforms to, and whether it conforms.	c	req.	partly	S	–	✓
G-32†	State whether a resource is a dataset, a dataset series or a service. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.3)*						
G-32.1	Users can know whether a resource is a dataset, a dataset series or a service.	c	req.	partly	S	–	✓
G-33†	State the type of a spatial data service, e.g. discovery, view or download. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.9)*						
G-33.1	Users can know the type of a spatial data service, e.g. discovery, view or download.	c	req.	beyond	S	–	✓
G-34†	Characterise each resource with keywords and name the vocabularies they come from. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.8)*						
G-34.1	Users can know the keywords that characterise each resource and the vocabularies they come from.	c	req.	partly	S	–	✓
G-35†	Link a service to the datasets or dataset series it operates on. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.6)*						
G-35.1	Users can find the datasets or dataset series a service operates on.	c	req.	beyond	S	–	✓
G-36†	Give each resource a unique identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.5)*						
G-36.1	Users can know the unique identifier of each resource.	c	req.	restates	S	–	✓
G-37†	State the conditions for accessing and using a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.15)*						
G-37.1	Users can know the conditions for accessing and using a resource.	c	req.	restates	S	–	✓
G-38†	State any limitations on public access to a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.15)*						
G-38.1	Users can know any limitations on public access to a resource.	c	req.	restates	S	–	✓
G-39†	State how often a resource is maintained and updated. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.27)*						
G-39.1	Users can know how often a resource is maintained and updated.	c	req.	partly	S	–	✓
G-40†	Report the topological consistency of a dataset as a data quality result. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14; §5.3)*						
G-40.1	Users can know the topological consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓
G-41†	Report the conceptual and domain consistency of a dataset as data quality results. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14)*						
G-41.1	Users can know the conceptual consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓
G-41.2	Users can know the domain consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓
G-42†	Make spatial datasets, dataset series and services searchable on general data portals. *(GeoDCAT-AP 1.0 §1; §3)*						
G-42.1	The profile makes spatial datasets, dataset series and services searchable on general data portals.	p	req.	–	P	–	
G-43†	Allow ISO 19115/INSPIRE metadata records to be converted into the RDF representation; conversion back is not required. *(GeoDCAT-AP 1.0 §1.1)*						
G-43.1	The syntax binding supports the conversion of ISO 19115/INSPIRE metadata records into the RDF representation; conversion back is not required.	s	req.	–	U	–	
G-44†	Keep the RDF binding as simple as possible, reusing existing vocabularies and avoiding new terms. *(GeoDCAT-AP 1.0 §1.1)*						
G-44.1	The profile keeps its RDF syntax binding as simple as possible.	p	req.	–	U	–	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-45†	Reuse the published alignments between EuroVoc, GEMET and the INSPIRE themes. *(GeoDCAT-AP 1.0 §1.1)*						
G-45.1	The profile reuses the published alignments between EuroVoc, GEMET and the INSPIRE themes.	p	req.	–	U	–	
G-46†	Feed recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned. *(GeoDCAT-AP 1.0 §1.1)*						
G-46.1	The GeoDCAT-AP working group feeds recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned.	s	req.	–	N	–	

HealthDCAT-AP HealthData@EU Pilot D6.2 (2024); base DCAT-AP 3.0.0; development case

69 rows (16 published), 121 atoms, 89 scored (51 requirements), 13 restate the base, 59 stated (84 at least in part); pre-consolidation corpus: 5 (22)

H-1	Use stable, resolvable identifiers so metadata records and their relationships remain reliably identifiable across catalogues. *(D6.2 §5.2.1.1)*						
H-1.1	Users can access each dataset's metadata record through a persistent, dereferenceable identifier (an HTTP URI).	c	req.	tightens	P	P	✓
H-1.2	Users can access the datasets and external resources a dataset links to through persistent, dereferenceable URIs.	c	req.	tightens	S	U	✓
H-2	Represent catalogue relationships as machine-interpretable links that support a health-data knowledge graph. *(D6.2 §5.2.1.2)*						
H-2.1	The profile represents the relationships between datasets and other resources (publications, codes) as machine-interpretable links, so the catalogue forms a knowledge graph.	p	req.	–	S	N	
H-3	Have data holders review their dataset descriptions at least once a year for accuracy and currency, and keep track of when each description was last reviewed. *(D6.2 §5.2.1.3)*						
H-3.1	Data holders check at least once a year that their dataset descriptions are accurate and up to date.	s	req.	–	P	N	
H-3.2	Users can know when a dataset description was last reviewed, separately from when it was last modified, where catalogues manage the review.	c	prop.	beyond	N	N	✓
H-4	Make metadata sufficiently interpretable and informative to support machine-assisted search, including semantic and generative-AI use. *(D6.2 §5.2.2)*						
H-4.1	Users can find datasets with natural-language and AI-based search, through meaningful free-text descriptions (what, why, how, who).	c	req.	partly	P	S	u
H-4.2	Users can find datasets by querying their concepts and relationships semantically.	c	req.	beyond	S	N	✓
H-5	Allow dataset discovery to be refined using health categories, access bodies, holder types, personal-data categories, age ranges, population size, quality labels and coding information. *(D6.2 §5.2.3)*						
H-5.1	Users can filter datasets by health category in a category facet, using the controlled list of EHDS Art. 33 categories.	c	req.	beyond	P	P	✓
H-5.2	Users can filter datasets by health data access body.	c	req.	beyond	P	P	✓
H-5.3	Users can filter datasets by publisher type.	c	req.	extends	P	P	✓
H-5.4	Users can filter datasets by the categories of personal data they contain.	c	req.	beyond	P	P	✓
H-5.5	Users can filter population datasets by the minimum typical age of their population.	c	req.	beyond	S	P	✓
H-5.6	Users can filter population datasets by the maximum typical age of their population.	c	req.	beyond	S	P	✓
H-5.7	Users can filter datasets by population size.	c	req.	beyond	P	P	✓
H-5.8	Users can filter datasets by quality label.	c	req.	beyond	S	P	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-5.9	Users can filter datasets by the coding values they use.	c	req.	beyond	S	P	✓
H-6 Adapt displayed filters to whether the catalogue query concerns personal or non-personal data. *(D6.2 §5.2.3)*							
H-6.1	The catalogue adapts the displayed search filters to whether the user is looking for personal or non-personal data.	s	req.	–	U	N	
H-7 Use semantic annotations and shared coding vocabularies to make health descriptions interoperable and searchable. *(D6.2 §5.2.5)*							
H-7.1	Users can find the datasets about the same concept across the data space, through shared concepts with persistent identifiers.	c	req.	beyond	S	U	✓
H-7.2	Users can know the health themes of a dataset, as HTTP URIs from an authoritative vocabulary.	c	req.	beyond	P	N	✓
H-7.3	Users can know the coding systems a dataset uses, as HTTP URIs.	c	req.	beyond	S	N	✓
H-7.4	Users can know the code values a dataset uses, linked to external standards such as ICD-10.	c	req.	beyond	S	N	✓
H-8 Describe at least one safe sample distribution for non-public datasets so users can prepare analyses without accessing sensitive records. *(D6.2 §5.2.6)*							
H-8.1	Users can access at least one sample distribution (mock-up, synthetic or anonymised data) of every non-public dataset.	c	req.	tightens	P	S	✓
H-9 Accompany sample data with definitions of its variables in a data dictionary. *(D6.2 §5.2.6)*							
H-9.1	Users can access a data dictionary that defines all data elements of each sample distribution.	c	req.	beyond	P	N	✓
H-10 Allow applicants to identify the variables they need, supporting data minimisation. *(D6.2 §5.2.6 (pp. 43–44))*							
H-10.1	Data access applicants can list the specific variables they need, in line with data minimisation.	s	req.	–	S	N	
H-11 Provide usable descriptions of data structures and coding so datasets can be understood and processed consistently. *(D6.2 §5.2.6.1)*							
H-11.1	Users can know the data structures of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.2	Users can know the data formats of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.3	Users can know the vocabularies of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.4	Users can know the classification schemes of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.5	Users can know the taxonomies of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	P	N	✓
H-11.6	Users can know the code lists of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-12 Describe logical dataset subdivisions and their relationships where large holdings need to be partitioned. *(D6.2 §5.2.6.2)*							
H-12.1	Users can know the coherent logical ('virtual') datasets within a large data warehouse, each with its own metadata record.	c	req.	restates	U	U	✓
H-12.2	Users can know how a data source breaks down into collections and datasets, from the dataset descriptions.	c	def.	?	P	U	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-13 State intellectual-property and reuse conditions, including those on sample distributions. *(D6.2 §5.2.6.3)*							
H-13.1	Users can know the conditions for re-use (rights) of every distribution, including sample distributions.	c	req.	tightens	P	N	✓
H-13.2	Users can know which parts of a dataset are protected by intellectual property rights.	c	req.	extends	P	N	✓
H-13.3	Users can know why parts of a dataset need intellectual-property protection.	c	req.	extends	N	N	✓
H-13.4	Users can access a redacted data dictionary, when intellectual property rights prevent a sample subset.	c	prop.	beyond	N	N	✓
H-14 Link to descriptive analytics that help prospective users assess a dataset. *(D6.2 §5.2.7)*							
H-14.1	Users can access analytics resources (dashboards, reports, statistics APIs) to assess a dataset without accessing the raw data.	c	req.	beyond	P	S	✓
H-15 Expose quality annotations or labels that help assess fitness for reuse. *(D6.2 §5.2.8)*							
H-15.1	Users can know the quality or utility annotations (e.g. a quality label or certificate) of a dataset or distribution.	c	req.	beyond	S	N	✓
H-15.2	Users can know the motivation of each quality annotation.	c	req.	beyond	P	N	✓
H-15.3	Users can know the content of each quality annotation, as text or as a link to a resource.	c	req.	beyond	P	N	✓
H-15.4	The profile describes dataset quality with one standardised approach across EU data spaces, aligned with StatDCAT-AP and mobilityDCAT-AP (DQV quality annotations).	p	req.	–	P	N	
H-16 Describe the purpose of data collection and the legal basis for processing personal data. *(D6.2 §5.2.9)*							
H-16.1	Users can know the purpose for which the data of a dataset were collected.	c	req.	beyond	S	S	✓
H-16.2	Users can know the legal basis for processing the personal data in a dataset.	c	req.	beyond	S	N	✓
H-17† Describe each dataset's source, scope, main characteristics, the nature of its health data and the conditions for making the data available. *(D6.2 §4 (quoting EHDS Art. 55))*							
H-17.1	Users can know the source of each dataset.	c	req.	tightens	S	P	✓
H-17.2	Users can know the scope of each dataset.	c	req.	tightens	S	U	✓
H-17.3	Users can know the main characteristics of each dataset.	c	req.	restates	S	U	✓
H-17.4	Users can know the nature of the electronic health data in each dataset.	c	req.	beyond	S	N	✓
H-17.5	Users can know the conditions for making the data of a dataset available.	c	req.	tightens	S	N	✓
H-18† Indicate the legislation that applies to a dataset (e.g. the EHDS Regulation), so health catalogues can filter the datasets in its scope. *(D6.2 §4.2; §5.3.1.1; §5.3.1.3 (Applicable legislations))*							
H-18.1	Users can know the legislation that applies to each dataset (e.g. the EHDS Regulation), so catalogues can filter the datasets in its scope.	c	prop.	tightens	P	N	✓
H-19† Classify each dataset's access rights as public or non-public, so catalogues know whether access needs an application. *(D6.2 §5.3.1.3 (Access rights))*							
H-19.1	Users can know each dataset's access rights (public, restricted or non-public), so catalogues know whether access needs an application; for NSIP data the rights are only 'non-public' or 'restricted'.	c	prop.	tightens	P	U	✓
H-20† Provide at least one distribution for every dataset: an access URL or download link for public data, a landing page of the health data access body for non-public data. *(D6.2 §5.3.1.1; §5.3.1.3 (Access rights; Dataset distribution))*							
H-20.1	Users can access at least one distribution of every dataset.	c	prop.	tightens	P	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-20.2	Users can access a public dataset through an access URL or download link, and a non-public dataset through the landing page of the health data access body or equivalent authority.	c	prop.	tightens	P	N	✓
H-21†	Name a contact point for every dataset, so users can ask about it without working out whom to contact. *(D6.2 §5.3.1.3 (Contact point))*						
H-21.1	Users can know the contact point of every dataset, which acts as its helpdesk.	c	prop.	tightens	S	N	✓
H-22†	Identify the health data access body that acts as the access gateway for a non-public or personal dataset. *(D6.2 §5.3.1.3 (Health data access body))*						
H-22.1	Users can know the health data access body that acts as access gateway for every dataset with personal or non-public data.	c	prop.	beyond	P	N	✓
H-23†	Indicate whether a dataset contains personal data. *(D6.2 §5.3.1.3 (Data type); §5.3.1.4)*						
H-23.1	Users can know whether a dataset contains personal or non-personal data (data type).	c	prop.	extends	S	U	✓
H-24†	Give each dataset's geographical coverage, preferably from controlled vocabularies, as a filter for populations. *(D6.2 §5.3.1.3 (Geographical coverage))*						
H-24.1	Users can filter datasets by their geographical coverage, when considering populations.	c	prop.	tightens	S	P	✓
H-25†	Characterise each dataset with a few main keywords per language rather than long lists. *(D6.2 §5.3.1.3 (Keyword))*						
H-25.1	Users can find each dataset by a few main keywords per language (10–15) rather than long lists.	c	prop.	tightens	P	N	✓
H-26†	Describe a dataset's lineage: how it was created and modified, and by whom. *(D6.2 §5.3.1.3 (Provenance))*						
H-26.1	Users can know a dataset's lineage: how it was created and modified, and by whom.	c	prop.	tightens	S	S	✓
H-27†	Identify the publisher as the data holder accountable for making the dataset accessible. *(D6.2 §5.3.1.3 (Publisher))*						
H-27.1	Users can know the publisher of a dataset, as the data holder accountable for making it accessible.	c	prop.	tightens	S	N	✓
H-28†	Describe the nature and type of the data holder as context for judging a dataset's credibility. *(D6.2 §5.3.1.3 (Publisher note; Publisher type))*						
H-28.1	Users can know the nature of the data holder, as context for judging the credibility of the dataset.	c	prop.	beyond	P	N	✓
H-28.2	Users can know the publisher type of the data holder, from a controlled vocabulary.	c	prop.	tightens	S	N	✓
H-29†	Tag every dataset in a health catalogue with the EU data theme for health. *(D6.2 §5.3.1.3 (Theme); §5.3.1.4)*						
H-29.1	Users can find every dataset in a health catalogue by the EU data theme for health.	c	prop.	tightens	P	N	✓
H-30†	Identify the data models and other standards a dataset conforms to (e.g. OMOP), so users can find data ready for their analysis. *(D6.2 §4.1; §5.2.6.2.1; §5.3.1.3 (Conforms to))*						
H-30.1	Users can know the data model or other standard a dataset conforms to (e.g. OMOP), so they can find data ready for their analysis.	c	req.	tightens	S	N	✓
H-30.2	Users can use machine-interpretable identifiers (HTTP URIs) for the standards a dataset conforms to.	c	req.	tightens	S	N	✓
H-31†	Link to documentation that helps users understand the dataset. *(D6.2 §5.3.1.3 (Documentation))*						
H-31.1	Users can access documentation that helps them understand the dataset.	c	prop.	tightens	S	N	✓
H-32†	Link a dataset to related resources and publications that show how it has been used. *(D6.2 §5.2.1.2; §5.3.1.3 (Is referenced by; Related resource); §6.1)*						
H-32.1	Users can find the publications and other resources that reference a dataset.	c	prop.	tightens	S	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-32.2	Users can find the resources related to a dataset.	c	prop.	tightens	S	N	✓
H-33† Identify the datasets a dataset was derived from, and let datasets derived from it refer back to it. *(D6.2 §5.3.1.3 (Source); §6.1)*							
H-33.1	Users can know the datasets a dataset was derived from; research datasets derived from a health dataset cite its metadata identifier.	c	prop.	tightens	S	U	✓
H-34† Give a landing page that provides access to the dataset and further information. *(D6.2 §5.3.1.3 (Landing page))*							
H-34.1	Users can access the dataset and further information through a landing page.	c	prop.	tightens	S	N	✓
H-35† State the language of a dataset where applicable. *(D6.2 §5.3.1.3 (Language))*							
H-35.1	Users can know the language of a dataset, where applicable.	c	prop.	tightens	S	N	✓
H-36† Describe the population a dataset covers, where it contains one. *(D6.2 §5.3.1.3 (Population coverage))*							
H-36.1	Users can know the population a dataset covers, where it contains one.	c	prop.	beyond	S	S	✓
H-37† State the size of a dataset (number of records, number of unique individuals; byte size of each distribution), so users can judge its value. *(D6.2 §2.3.1.3 and §5.3.1.2 (DGA Art. 8); §5.3.1.3 (Number of records))*							
H-37.1	Users can know the number of records in a dataset.	c	prop.	beyond	S	P	✓
H-37.2	Users can know the number of unique individuals in a dataset.	c	prop.	beyond	S	P	✓
H-37.3	Users can know the byte size of each distribution.	c	prop.	tightens	S	U	✓
H-38† Give a dataset's temporal coverage, leaving the end open while collection continues. *(D6.2 §5.3.1.3 (Temporal coverage))*							
H-38.1	Users can know a dataset's temporal coverage, with the end left open while collection continues.	c	prop.	restates	S	N	✓
H-39† State a dataset's temporal resolution, to judge whether it can be reused. *(D6.2 §5.3.1.3 (Temporal resolution))*							
H-39.1	Users can know a dataset's temporal resolution, to judge whether it can be reused.	c	prop.	tightens	S	N	✓
H-40† State when a dataset must be deleted (retention period), and keep its record available afterwards. *(D6.2 §5.3.1.3 (Retention period))*							
H-40.1	Users can know the retention period after which a dataset is to be deleted.	c	prop.	beyond	P	N	✓
H-40.2	The catalogue keeps a deleted dataset's metadata record available, because publications can refer to it.	s	prop.	–	U	U	
H-41† Give a version designation where the holder follows a versioning strategy. *(D6.2 §5.3.1.3 (Version))*							
H-41.1	Users can know the version designation of a dataset, where the holder follows a versioning strategy.	c	prop.	restates	S	N	✓
H-42† State the spatial resolution where it applies, mainly for geospatial data. *(D6.2 §5.3.1.3 (Spatial resolution))*							
H-42.1	Users can know the spatial resolution of a dataset where it applies, mainly for geospatial data.	c	prop.	restates	S	N	✓
H-43† Record when each catalogue entry was listed and last changed, and which original metadata it was created from. *(D6.2 §5.2.1.1)*							
H-43.1	Users can know when each catalogue entry was listed.	c	req.	restates	S	N	✓
H-43.2	Users can know when each catalogue entry was last modified.	c	req.	restates	S	N	✓
H-43.3	Users can know the original metadata an aggregated catalogue entry was created from.	c	req.	restates	S	U	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-44†	State which application profile a metadata record follows, so aggregators can validate it. *(D6.2 §5.2.1.3)*						
H-44.1	Users can know which application profile a metadata record follows, so aggregators can validate it.	c	req.	restates	S	U	✓
H-45†	Indicate whether a dataset's holder is a trusted data holder, as a search facet. *(D6.2 §5.2.3 (facet list))*						
H-45.1	Users can filter datasets by whether their holder is a trusted data holder, in a search facet.	c	prop.	beyond	U	U	✓
H-46†	Provide the information the EHDS data quality and utility label needs: documentation, maturity of quality management, coverage, access and provision times, and data modifications. *(D6.2 §5.2.8 (quoting EHDS Art. 56))*						
H-46.1	Users can know, for the EHDS data quality and utility label, how representative the sampled population is, when applicable.	c	req.	beyond	S	P	✓
H-46.5	Users can know, for the EHDS data quality and utility label, the average time a natural person appears in the dataset.	c	req.	beyond	S	P	✓
H-46.2	Users can know, for the EHDS data quality and utility label, the maturity of the data quality management processes, including review and audit processes and the examination of biases.	c	req.	beyond	S	N	✓
H-46.3	Users can know, for the EHDS data quality and utility label, the time between collecting the data and adding them to the dataset.	c	req.	beyond	S	N	✓
H-46.6	Users can know, for the EHDS data quality and utility label, the time to provide the data after a data access application is approved.	c	req.	beyond	S	N	✓
H-46.4	Users can know, for the EHDS data quality and utility label, the modifications of the data: merging and adding data to an existing dataset, including links with other datasets.	c	req.	beyond	S	P	✓
H-47†	Standardise the minimum metadata elements that health data holders must provide under the EHDS Regulation. *(D6.2 §2.2)*						
H-47.1	The profile standardises the minimum metadata elements that health data holders are required to provide under the EHDS Regulation.	p	req.	–	S	N	
H-48†	Remain fully compatible with DCAT-AP. *(D6.2 §2.2)*						
H-48.1	The profile remains fully compatible with DCAT-AP.	p	req.	–	S	N	
H-49†	Align the minimum elements for open health data with those of DCAT-AP for High-Value Datasets. *(D6.2 §4.2; §5.3.1.1)*						
H-49.1	The profile aligns the minimum elements for open health data with those of DCAT-AP for High-Value Datasets.	p	prop.	–	P	N	
H-50†	Align the minimum elements for non-open, non-personal health data with the metadata of national single information points under the Data Governance Act. *(D6.2 §5.3.1.2)*						
H-50.1	The profile aligns the minimum elements for non-open, non-personal health data with the NSIP metadata of the Data Governance Act.	p	prop.	–	P	N	
H-51†	Define stricter, conditional minimum element sets for protected and for sensitive (personal) health data. *(D6.2 §4.2; §5.3.1.4)*						
H-51.1	The profile defines stricter, conditional minimum element sets for protected and for sensitive (personal) health data.	p	req.	–	P	N	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-52†	Take coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata). *(D6.2 §5.2.4; §5.2.5)*						
H-52.1	The profile takes coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata).	p	req.	–	P	U	
H-53†	Publish data dictionaries in one harmonised machine-readable format (CSVW), so variable definitions are consistent across catalogues. *(D6.2 §5.2.6.1)*						
H-53.1	The profile provides for data dictionaries in one harmonised machine-readable format (CSVW).	p	req.	–	P	N	
H-54†	Provide descriptions in at least one official EU language, preferably the holder's native language, with language tags. *(D6.2 §6.1.3 (quoting EHDS Art. 55(1a)))*						
H-54.1	The profile provides for dataset descriptions in at least one official EU language.	p	req.	–	P	N	
H-54.2	The profile encourages data holders to create metadata in their native language.	p	req.	–	P	N	
H-55†	Expose national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest. *(D6.2 §2.3.1.3; §6.2)*						
H-55.1	Health data access bodies expose the national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest.	s	req.	–	S	P	
H-56†	Keep the data holder's record as the single source of truth when aggregating: copies refer to the original, and access bodies add a persistent identifier without replacing the original one. *(D6.2 §5.2.1.1)*						
H-56.1	Aggregating catalogues keep their copies of metadata linked to the original record at its source.	s	req.	–	S	P	
H-56.2	Data holders update their metadata only at its primary source; they, not health data access bodies, own and edit it.	s	req.	–	P	U	
H-56.3	Health data access bodies assign an additional persistent URI to a record whose identifier does not meet the profile's identifier rules, keeping the data holder's original identifier.	s	req.	–	P	P	
H-57†	Detect broken URIs in the links between catalogue resources. *(D6.2 §5.2.1.2)*						
H-57.1	Those operating the health data space can detect broken URIs in the links between catalogue resources.	s	req.	–	U	N	
H-58†	Measure the maturity of the health data space from its metadata, such as the share of URIs, the level of interoperability and conformity to standards. *(D6.2 §5.2.1.2)*						
H-58.1	Those governing the health data space measure its maturity from its metadata (share of URIs, interoperability, conformity to standards).	s	req.	–	P	N	
H-59†	Order search facets by their relevance, as shown in the pilot use cases. *(D6.2 §5.2.3)*						
H-59.1	The search interface orders facets by their relevance, as shown in the pilot use cases.	s	req.	–	N	U	
H-60†	Let applicants attach an analysis script to a data request, to obtain anonymised statistical answers. *(D6.2 §5.2.6 (EHDS Art. 47))*						
H-60.1	Applicants can attach an analysis script to a data request, to obtain anonymised statistical answers.	s	req.	–	P	U	
H-61†	Support data holders in transforming existing metadata into the profile, with a user-friendly multilingual editor that needs no technical expertise. *(D6.2 §6; §6.1)*						
H-61.1	Data holders get support in transforming their existing metadata into the profile.	s	req.	–	S	P	
H-61.2	Data holders get a user-friendly, multilingual metadata editor that needs no technical expertise.	s	req.	–	P	P	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-62†	Validate metadata against the profile (mandatory elements, data types, controlled vocabularies) before publication and when aggregating. *(D6.2 §6.1.2; §6.2.1)*						
H-62.1	Data holders and metadata aggregators validate metadata against the profile (mandatory elements, data types, controlled vocabularies): data holders before publication, aggregators when aggregating.	s	req.	–	S	U	
H-63†	Assign responsible metadata managers at data holders and access bodies; the data holder, not the access body, owns and edits the metadata. *(D6.2 §6.1)*						
H-63.1	Data holders and health data access bodies each designate a responsible metadata manager.	s	req.	–	P	N	
H-64†	Let access bodies enrich metadata automatically when indicators show that a dataset falls under the EHDS. *(D6.2 §5.3.1.1)*						
H-64.1	Health data access bodies enrich metadata automatically (e.g. add the EHDS as applicable legislation) when indicators show a dataset falls under the EHDS.	s	prop.	–	U	U	
H-65†	Set up revision and maintenance procedures, with clear responsibility, for the profile. *(D6.2 §6.2)*						
H-65.1	The profile's maintainers set up revision and maintenance procedures, with clear responsibility.	s	prop.	–	U	N	
H-66†	Provide the elements DCAT-AP already makes mandatory: a title and a description for every dataset and an access URL for every distribution. *(D6.2 §5.3.1.1; §5.3.1.2 (NSIP table); §5.3.1.3 (Title; Description))*						
H-66.1	Users can know the title of every dataset.	c	prop.	restates	S	N	✓
H-66.2	Users can know the description of every dataset.	c	prop.	restates	S	N	✓
H-66.3	Users can access the data of each distribution, or information on how to request it, through an access URL.	c	prop.	restates	S	N	✓
H-67†	Give each distribution of protected (NSIP) data exactly one file format, and offer other formats as separate distributions. *(D6.2 §5.3.1.2 (NSIP distribution table))*						
H-67.1	Users can know the file format of each distribution, with exactly one format per distribution for protected (NSIP) data.	c	prop.	tightens	P	N	✓
H-67.2	Users can access the same data in another format as a separate distribution.	c	prop.	restates	S	N	✓
H-69†	Offer a common harmonised data validator in the infrastructure to check datasets that follow common data models. *(D6.2 §5.2.6.2.1)*						
H-69.1	The infrastructure offers a common harmonised data validator that checks the quality and consistency of datasets that follow common data models.	s	prop.	–	P	N	
H-68†	State how often a dataset is updated (frequency). *(D6.2 §5.3.1.3 (recommended properties: frequency))*						
H-68.1	Users can know how often a dataset is updated (frequency).	c	prop.	tightens	S	N	✓

StatDCAT-AP StatDCAT-AP 1.0.0 (2016); base DCAT-AP 1.1; development case
22 rows (7 published), 27 atoms, 14 scored (10 requirements), 6 restate the base, 4 stated (8 at least in part)

S-1	Expose the dimensions that distinguish observations, including time and demographic or regional dimensions. *(StatDCAT-AP 1.0 §7.2.1)*						
S-1.1	Users can know the dimensions that identify observations in a statistical dataset (e.g. time, sex, age, region), described the same way whatever the distribution format.	c	req.	beyond	P	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
S-2 Expose attributes needed to interpret observations, such as scaling and observation status. *(StatDCAT-AP 1.0 §7.2.1)*							
S-2.1	Users can know the attributes needed to qualify and interpret observed values (e.g. units of measure, scaling factors, observation status), described the same way whatever the distribution format.	c	req.	beyond	P	–	✓
S-3 Associate statistical datasets with descriptive quality information. *(StatDCAT-AP 1.0 §7.2.2)*							
S-3.1	Users can know the quality information of a dataset or distribution.	c	req.	beyond	N	–	✓
S-3.2	Users can know the motivation of each quality annotation.	c	req.	beyond	N	–	✓
S-3.3	Users can know the quality information, as text or as a link to quality information published elsewhere.	c	req.	beyond	N	–	✓
S-4 Identify delivery resources that present dataset visualisations. *(StatDCAT-AP 1.0 §7.2.3)*							
S-4.1	Users can access visualisations of the data (tables, graphs or interactive views).	c	req.	extends	N	–	✓
S-5 Indicate the actual number of statistical series represented in a dataset. *(StatDCAT-AP 1.0 §7.2.4)*							
S-5.1	Users can know the actual number of data series in a dataset, as distinct from the number of observations and the theoretical number of combinations.	c	req.	beyond	N	–	✓
S-6 Identify the unit in which dataset observations are measured. *(StatDCAT-AP 1.0 §7.2.5)*							
S-6.1	Users can know the unit of measurement of the observations in a dataset.	c	req.	beyond	S	–	✓
S-7 Represent time-series coverage and periodicity without relying on an ambiguous shorthand. *(StatDCAT-AP 1.0 §7.2.6)*							
S-7.1	Users can know the start of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.2	Users can know the end of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.3	Users can know the periodicity of a time series.	c	guid.	restates	U	–	✓
S-8† Improve the discoverability of statistical datasets on general open data portals. *(StatDCAT-AP 1.0.0 §2.2 (charter); §5.1)*							
S-8.1	The profile improves the discoverability of statistical datasets on general open data portals.	p	req.	–	S	–	
S-9† Let statistical portals feed general open data portals by exporting their metadata in the profile, with guidance for SDMX-based systems. *(StatDCAT-AP 1.0.0 §5.2; §8; Annexes IV–V)*							
S-9.1	Statistical portals can export their metadata in the profile, so general open data portals can take it up.	s	req.	–	S	–	
S-9.2	Organisations with SDMX-based systems get guidance for mapping and extracting their metadata into the profile.	s	req.	–	U	–	
S-10† Conform fully to DCAT-AP 1.1, meeting every obligation of its conformance statement. *(StatDCAT-AP 1.0.0 §7.1)*							
S-10.1	The profile conforms fully to DCAT-AP and keeps statistical descriptions aligned with DCAT and DCAT-AP.	p	req.	–	P	–	
S-11† Concentrate on metadata elements that help users discover datasets. *(StatDCAT-AP 1.0.0 §6.2)*							
S-11.1	The profile concentrates on the metadata elements that help users discover datasets.	p	req.	–	P	–	
S-12† Take terms from existing, well-known and well-maintained vocabularies, including the ISA Core Vocabularies and Eurostat's metadata vocabularies. *(StatDCAT-AP 1.0.0 §6.2)*							
S-12.1	The profile takes its terms from existing, well-known and well-maintained vocabularies.	p	req.	–	S	–	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
S-13†	Prefer common controlled vocabularies maintained by the EU Publications Office. *(StatDCAT-AP 1.0.0 §6.2)*						
S-13.1	The profile prefers common controlled vocabularies, preferably those maintained by the EU Publications Office.	p	req.	–	P	–	
S-14†	Balance simplicity and expressiveness for a wide, non-specialist audience. *(StatDCAT-AP 1.0.0 §6.2)*						
S-14.1	The profile balances simplicity and expressiveness for a wide, non-specialist audience.	p	req.	–	N	–	
S-15†	Make the profile usable for datasets in any format, such as SDMX, Data Cube or CSV. *(StatDCAT-AP 1.0.0 §7.1; Annex III)*						
S-15.1	The profile is usable for datasets in any format, such as SDMX, Data Cube or CSV.	p	req.	–	S	–	
S-16†	Classify statistical datasets by EU data theme, mapping Eurostat's own themes onto it. *(StatDCAT-AP 1.0.0 §7.8.3)*						
S-16.1	Users can find statistical datasets by EU data theme.	c	req.	restates	P	–	✓
S-17†	Link an RDF distribution to the data structure definition it follows. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-17.1	Users can know which data structure definition an RDF distribution follows.	c	guid.	restates	S	–	✓
S-18†	Record publication and modification times to the time of day when data are published more than once a day. *(StatDCAT-AP 1.0.0 §11; Annex III)*						
S-18.1	Users can know the publication and modification times of a dataset to the time of day, when its data are published more than once a day.	c	req.	restates	P	–	✓
S-19†	Identify the measures a statistical dataset contains, as SDMX and Data Cube define them. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-19.1	Users can know the measures a statistical dataset contains, as SDMX and Data Cube define them.	c	def.	beyond	S	–	
S-20†	Describe the quality of statistical data in more detail, e.g. following the ESS Single Integrated Metadata Structure (SIMS). *(StatDCAT-AP 1.0.0 §7.2.2; Annex III)*						
S-20.1	Users can know the quality of statistical data in more detail.	c	def.	beyond	N	–	
S-21†	Show how DDI descriptions of datasets can be expressed with the profile. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-21.1	The profile shows how DDI descriptions of datasets can be expressed with it.	p	def.	–	U	–	
S-22†	Document, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate. *(StatDCAT-AP 1.0.0 Annex III; Annex VII)*						
S-22.1	The profile documents, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate.	p	req.	–	P	–	

DCAT-AP-JRC Perego et al., SDSVoc 2016, paper 27; base DCAT-AP 1.1; development case

11 rows (5 published), 18 atoms, 17 scored (9 requirements), 4 restate the base, 6 stated (12 at least in part)

JRC-1	Identify the people who created a dataset. *(Metadata elements relevant for scientific data, numbered item 1)*						
JRC-1.1	Users can know the authors of a dataset.	c	req.	beyond	S	–	✓
JRC-2	Describe the origin and processing history of data. *(Metadata elements relevant for scientific data, numbered item 2)*						
JRC-2.1	Users can know the lineage of a dataset.	c	req.	restates	S	–	✓
JRC-3	Give prospective users practical guidance on using a dataset. *(Metadata elements relevant for scientific data, numbered item 3)*						
JRC-3.1	Users can know how to use the data of a dataset (usage notes).	c	req.	beyond	U	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
JRC-4	Connect datasets to associated scientific publications. *(Metadata elements relevant for scientific data, numbered item 4)*						
JRC-4.1	Users can find the scientific publications about a dataset.	c	req.	beyond	P	–	✓
JRC-5	Identify source datasets used in producing a dataset. *(Metadata elements relevant for scientific data, numbered item 5)*						
JRC-5.1	Users can find the input data of a dataset (the data used to create it).	c	req.	restates	P	–	✓
JRC-6†	Give datasets persistent identifiers, encoded as HTTP URIs where possible, and tell primary identifiers (e.g. DOIs) from alternative ones. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Identifiers")*						
JRC-6.1	Users can reliably filter and link datasets by their authors and input data, through persistent identifiers.	c	req.	tightens	P	–	✓
JRC-6.2	Users can use identifiers encoded as HTTP URIs, wherever possible.	c	open	tightens	U	–	✓
JRC-6.3	Users can know which of a dataset's identifiers is the primary one (e.g. a DOI) and which are alternative ones.	c	req.	restates	S	–	✓
JRC-6.4	Users can cite a dataset by its DOI, when DataCite citations are generated.	c	req.	tightens	S	–	✓
JRC-6.5	Users can know the scheme or type of each identifier.	c	req.	restates	S	–	✓
JRC-7†	Express the roles that agents play for a dataset beyond publisher and contact point, consistently across catalogues. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Agent roles")*						
JRC-7.1	Users can know the roles that agents play for a dataset beyond publisher and contact point, from a minimal, consistent set of unambiguous roles.	c	open	beyond	P	–	✓
JRC-8†	Indicate when a distribution gives access through a service or API rather than to the data itself, and describe the service interface with the query parameters needed to select data. *(Perego et al. 2016 (SDSVoc paper 27), "Modelling service/API-based data access")*						
JRC-8.1	Users can know when a distribution points to a service or API rather than directly to the data.	c	open	extends	S	–	✓
JRC-8.2	Users can know the API or service interface and the query parameters needed to select data, so software agents can use it directly.	c	open	beyond	P	–	✓
JRC-9†	Describe provenance in enough detail to support reproducibility and to track how data are used. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (detailed specification of data provenance))*						
JRC-9.1	Users can know the provenance of a dataset in enough detail to support reproducibility and to track how the data are used.	c	open	beyond	P	–	✓
JRC-10†	Describe data and metadata quality assessments and include users' feedback on the data. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (modelling and using data quality assessments))*						
JRC-10.1	Users can know the data quality assessments of a dataset.	c	open	beyond	U	–	✓
JRC-10.2	Users can know the feedback users have given on the data, as quality information.	c	open	beyond	U	–	✓
JRC-10.3	Users can know the quality assessments of the metadata of a dataset.	c	open	beyond	N	–	✓
JRC-11†	Publish metadata on the Web so that the data are visible and discoverable through search engines. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (optimising metadata publication on the Web))*						
JRC-11.1	The catalogue publishes its metadata on the Web, so that the data are visible and discoverable through search engines.	s	open	–	P	–	